U-space Geoawareness Static Data Provision SWIM Service Description Document

General Service Description	
Service description identification **SWIM-SERV-030**	**Title:** U-space Static Geoawareness Data Provision SWIM Service Description Document **Edition:** V1.1 **Reference Date:** 2026-06-23
Service identification **SWIM-SERV-040**	**Name:** U-space Static Geoawareness Data Provision **Version:** V1.0
Service abstract **SWIM-SERV-050**	The service provides authoritative data for UAS Geographical Zones (UGZ) to U-space Service Providers (USSPs) from the Common Information System (CIS). The data describes non-aeronautical zones in which the operation of unmanned aircraft is prohibited, restricted, or subject to conditions. Aeronautical data and dynamic restrictions are provided by Skyguide and described in a distinct SDD available on the Swiss FOCA U-Space repository.
Service Provider and Contact Information	
Service Provider **SWIM-SERV-060**	**Name:** Common Information Service (FOCA and Skyguide) **Description of organization:** The Common Information Service refers to an organisation that is certified by a competent authority to provide U-space services in the U-space airspace (Ref: GM1a Article 7, U-space service providers). In the Swiss context, the Common Information Service is operated jointly by the Federal Office of Civil Aviation (FOCA) and Skyguide. **Contact information:** - Email: gis@bazl.admin.ch and rpas@bazl.admin.ch. **Support channel:**

	- Alerts and notifications: Channel #cis-geoawareness on Slack - Issue tracker: https://github.com/focagis/geoawareness-cis/issues
Service Characteristics	
Geographical extent of information **SWIM-SERV-090**	Switzerland and Liechtenstein. The data set covers Swiss UAS Geographical Zones.
Service categories **SWIM-SERV-100**	**Service category:** Common Information Service (U-Space)
Service lifecycle information **SWIM-SERV-110**	**Lifecycle stage:** Prospective
Service standard reference **SWIM-SERV-120**	**Service standard reference:** - EUROCAE Standard ED-318, edition 2024 **Implemented options of the service standard:** - The UGZ data model, structure and file format follow the Swiss UAS Geodata model in ED-318, including the conversion from ED-269.
High-level Description of Service Offer	
Operational environment **SWIM-SERV-130**	The service addresses the regulatory need to provide authoritative UAS Geographical Zone data to U-space Service Providers operating in Swiss U-space airspace. The legal basis for the service is summarised below. - Swiss Ordinance of the DETEC on special category aircraft - Art. 29e, par. 1 - Commission Implementing Regulation (EU) 2021/664 - Art. 5, par. 1 - Commission Implementing Regulation (EU) 2019/947 - Art. 15, par. 3
Service functions **SWIM-SERV-140**	The service makes UAS Geographical Zone data available as part of the common information services of the U-space airspace, in accordance with Article 5(1) of Commission Implementing Regulation

	(EU) 2021/664. It provides static airspace restrictions to U-space service providers in EUROCAE ED-318 format over HTTPS, enabling them to determine the geographical zones and conditions applicable to a UAS operation. Non-aeronautical data are expressed as UAS Geographical Zones (UGZ). **Data types:** - 5 km airport zones - Perimeter of prisons - Military installations - Outdoor switchgear / substations at network level 2 of power-supply facilities - Ruswil compressor stations and Wallbach measuring station (natural gas supply facilities) - Nuclear power plants and interim storage facility in Würenlingen - Swiss National Park - Water and migratory bird reserves - Federal hunting-ban areas - Vaduz Castle - Cantonal protected areas
Limitations and Constraints on Using the Service	
Service access and use conditions **SWIM-SERV-150**	The CIS provides the dataset via the HTTPS file-transfer protocol: focagis.github.io/geoawareness-cis/ed318.json If there are errors in a retrieved file, the user is expected to try again using the latest valid file update provided in the meantime.
Security constraints **SWIM-SERV-160**	Data is publicly available.
Quality Aspects	
Quality of service **SWIM-SERV-180**	**Timeliness** Refreshed every 6 hours, at 05:15, 11:15, 17:15 and 23:15 LT. **Availability**

	Best effort **Validity** Data effective dates follow the ED-318 standard: data is valid from validFrom until the next update (eg. validFrom + 6 hours). In case of file errors, availability is maintained by allowing users to fall back to the latest valid file update.
Service validation information **SWIM-SERV-200**	Service will be verified using the InterUSS Automated Test Baseline prior to software deployment using a test configuration with a test baseline identifier of XXX. (To be provided in next version of the SDD)
Behavior of the Service	
Application message exchange pattern **SWIM-SERV-210**	Data is retrieved as a file over HTTPS (file-pull)
Service behavior **SWIM-SERV-220**	The service publishes authoritative UAS Geographical Zone data describing the zones in which the operation of unmanned aircraft is prohibited, restricted, or subject to conditions. Each data set is generated and made available on a fixed refresh cycle (see Quality of service, SWIM-SERV-180) and remains valid from its `validFrom` date until it is superseded by the next scheduled update, in accordance with the ED-318 validity model. The data is served over a HTTPS endpoint; where a published file contains errors, the most recent valid file remains available until the next update.
Service Implementation and Structural Details	
Service interfaces **SWIM-SERV-240**	Endpoint serving the Swiss national UGZ dataset over HTTPS.
Service interface protocols and data format **SWIM-SERV-260**	**Protocol:** HTTPS file transfer. **Data format:** ED-318 as defined in the Swiss UAS Geodata model, serialised as JSON. Data is converted from ED-269.
Service operations **SWIM-SERV-270**	**Retrieve Static Geo-Awareness data** Method: HTTP GET Request: No parameters

	Response: - Code: 200 OK - Body: ED-318 JSON UGZ dataset
Service messages **SWIM-SERV-280**	Two messages, a request and a response, see SWIM-SERV-270. Expected error responses are standard HTTP responses (especially codes 500 and 502).
Information Aspects of the Service	
Information definition (minimum) **SWIM-SERV-290**	The information is structured as UAS Geographical Zones (UGZ) following the Swiss UAS Geodata model. Each zone may carry one or more restriction codes that define the operational effect within the zone. The restriction codes are defined in Annex A and operating rules in Annex B.
Resources	
Machine-readable service interface definition **SWIM-SERV-320**	The machine-readable data is published as ED-318 JSON at the following endpoint: https://focagis.github.io/geoawareness-cis/ed318.json
Abbreviations and acronyms **SWIM-SERV-350**	- API: Application Programming Interface - CIS: Common Information Service - CTR: Control Zone - DETEC: Federal Department of the Environment, Transport, Energy and Communications (Switzerland) - DSS: Discovery and Synchronization Service - ED: EUROCAE Document - EU: European Union - FOCA: Federal Office of Civil Aviation (Switzerland) - HTTPS: Hypertext Transfer Protocol Secure - LT: Local Time - OR: Operating Rule - REC: Restriction (code) - SDD: Service Description Document - SFC: Surface (altitude reference) - UAS: Unmanned Aircraft System - UGZ: UAS Geographical Zone - USSP: U-space Service Provider

Annex A: Restriction codes

Reference: Swiss UAS Geodata model.

Restriction ID	Description
REC01	The operation of unmanned aircraft is prohibited.
REC02a	The operation of unmanned aircraft is only allowed with an exemption permit.
REC02b	The operation of unmanned aircraft weighing more than 250 g is only allowed with an exemption permit.
REC02c	The operation of unmanned aircraft weighing more than 250 g is only permitted from an altitude of 120 m above ground with an exemption permit.
REC03	The operation of unmanned aircraft is permitted under certain conditions.
REC04	The operation of unmanned aircraft is permitted under certain conditions.*
REC05	The operation of unmanned aircraft is permitted.

* Duplicated on-purpose

Annex B: Operating Rules

Reference: Swiss UAS Geodata model.

Rule ID	Description
OR010	Operation of unmanned aircraft weighing less than or equal to 250 g.
OR011	Operation of unmanned aircraft weighing more than 250 g.
OR020	Operation of unmanned aircraft below or at an altitude of 120 m above ground.
OR021	Operation of unmanned aircraft above an altitude of 120 m above ground.